# Visual roadmaps

Three routes through the same lessons. The quick path covers everything in order; the other two put either model training or AI applications first. Effort ranges are study estimates.

## Quick path: complete grip roadmap

The full sequence in the shortest sensible order. Each step builds on the one before. Finish with a capstone that combines ML and AI.

1

### The big picture and your setup

4–6 h

What AI and ML are, the project lifecycle, Python and .NET on your laptop.

- What AI, machine learning, deep learning and generative AI are **Beginner** 60 min
- How an AI or ML project runs from idea to production **Beginner** 75 min
- Set up your laptop for AI and ML in Python and .NET **Beginner** 1.5 h

2

### Data skills

8–12 h

Data types, collection, labelling, cleaning, exploring and features.

- Data basics: types, features, labels and splits **Beginner** 60 min
- Collecting and labelling data, with SQL **Beginner** 1.5 h
- Cleaning and exploring data **Beginner** 1.5 h
- Feature engineering: turning raw data into model inputs **Intermediate** 1.5 h

3

## Core machine learning

12–18 h

How models learn, regression, classification, the main algorithms and unsupervised learning.

- How a model learns: loss, training and overfitting **Beginner** 75 min
- Regression: predicting a number **Beginner** 75 min
- Classification: picking a category **Beginner** 75 min
- The main ML algorithms and when to use each **Intermediate** 1.5 h
- Unsupervised learning: groups, compression and anomalies **Intermediate** 75 min

4

## Measure, improve and apply ML

8–12 h

Metrics, tuning, explanations, forecasting and recommenders.

- Measuring models honestly **Intermediate** 1.5 h
- Improving and explaining models **Intermediate** 1.5 h
- Time-series forecasting **Intermediate** 75 min
- Recommender systems **Intermediate** 75 min

5

## Deep learning

14–20 h

Neural networks, training, vision, text, transformers, generation, speech, RL and graphs.

- Neural networks in plain words **Intermediate** 1.5 h
- Training deep networks well **Intermediate** 1.5 h
- Computer vision: classification, detection and segmentation **Intermediate** 1.5 h
- Text and sequences: tokens, embeddings and RNNs **Intermediate** 75 min
- Transformers and attention **Intermediate** 1.5 h
- Generative models: autoencoders, GANs and diffusion **Advanced** 75 min
- Speech and audio: recognition and text-to-speech **Advanced** 60 min
- Reinforcement learning **Advanced** 75 min
- Graph neural networks: learning from connections **Advanced** 60 min

6

## Large language models

14–20 h

How LLMs work, design, train on a laptop, scale up, fine-tune, run locally, evaluate.

- How large language models work **Intermediate** 75 min
- Tokens and sampling settings **Intermediate** 60 min
- Designing a language model **Advanced** 75 min
- Train a small language model on your laptop **Advanced** 2 h
- From laptop to big hardware: scaling training **Expert** 1.5 h
- Fine-tuning: SFT, LoRA, QLoRA and preference tuning **Advanced** 1.5 h
- Running language models on your own machine **Intermediate** 75 min
- Evaluating language models **Advanced** 60 min

7

## Talk to models and add knowledge

10–14 h

LLM APIs, prompting, embeddings, RAG and advanced retrieval.

- Calling an LLM from your application **Beginner** 75 min
- Prompt engineering that works in production **Beginner** 60 min
- Embeddings and vector search **Intermediate** 75 min
- RAG: answering from your own documents **Intermediate** 2 h
- Advanced RAG: chunking, hybrid search, re-ranking and evaluation **Advanced** 1.5 h

8

## Agents, multimodal and trust

12–16 h

Tools, agents, frameworks, multimodal apps, evaluation, security and responsible AI.

- Tools and agents with human approval **Advanced** 2 h
- AI frameworks and MCP: what to use in Python and .NET **Intermediate** 60 min
- Multimodal AI apps: images, documents and voice **Intermediate** 60 min
- Evaluating AI applications **Intermediate** 75 min
- Security for AI applications **Advanced** 75 min
- Responsible AI: fairness, privacy, transparency and oversight **Intermediate** 60 min
- "Generative AI and Agentic AI: the whole picture" **Advanced** 60 min

9

**Production** 10–14 h

Serving, MLOps, monitoring, cost and scale, architectures.

- Serving a model as an API **Intermediate** 1.5 h
- "MLOps: versioning, pipelines and safe releases" **Advanced** 1.5 h
- Monitoring models and AI apps in production **Advanced** 75 min
- Speed, cost and scaling **Advanced** 75 min
- Production architecture for AI systems **Expert** 2 h

10

**Capstones and real cases** 30–60 h

Use the decision guide on real cases and build at least two capstones.

- Capstone 1: Customer churn prediction service 20–30 h
- Capstone 4: Company knowledge assistant (RAG) 20–35 h
- Capstone 6: Train a small language model on your laptop 25–40 h
- Capstone 8: Fine-tune a small model with LoRA 20–30 h

# ML engineer view roadmap

Focus on training models from data and running them in production. Read Part 6 later for LLM applications.

1

**Big picture and setup** 4–6 h

Lifecycle and tools.

- What AI, machine learning, deep learning and generative AI are **Beginner** 60 min
- How an AI or ML project runs from idea to production **Beginner** 75 min
- Set up your laptop for AI and ML in Python and .NET **Beginner** 1.5 h

2

## Data

8–12 h

Collect, clean, explore, engineer features.

- Data basics: types, features, labels and splits **Beginner** 60 min
- Collecting and labelling data, with SQL **Beginner** 1.5 h
- Cleaning and exploring data **Beginner** 1.5 h
- Feature engineering: turning raw data into model inputs **Intermediate** 1.5 h

3

## Supervised learning

8–12 h

How models learn, regression and classification.

- How a model learns: loss, training and overfitting **Beginner** 75 min
- Regression: predicting a number **Beginner** 75 min
- Classification: picking a category **Beginner** 75 min

4

## Algorithms and unsupervised learning

6–10 h

Which algorithm when; clustering, PCA and anomalies.

- The main ML algorithms and when to use each **Intermediate** 1.5 h
- Unsupervised learning: groups, compression and anomalies **Intermediate** 75 min

5

## Evaluation and improvement

6–10 h

Metrics, validation, tuning, explanations.

- Measuring models honestly **Intermediate** 1.5 h
- Improving and explaining models **Intermediate** 1.5 h

6

### Forecasting and recommendations

6–10 h

Two of the most common business uses.

- Time-series forecasting **Intermediate** 75 min
- Recommender systems **Intermediate** 75 min

7

### Deep learning

14–20 h

Networks, training, vision and other model families.

- Neural networks in plain words **Intermediate** 1.5 h
- Training deep networks well **Intermediate** 1.5 h
- Computer vision: classification, detection and segmentation **Intermediate** 1.5 h
- Text and sequences: tokens, embeddings and RNNs **Intermediate** 75 min
- Transformers and attention **Intermediate** 1.5 h
- Reinforcement learning **Advanced** 75 min

8

### MLOps and production

10–14 h

Serve, track, monitor and scale models.

- Serving a model as an API **Intermediate** 1.5 h
- "MLOps: versioning, pipelines and safe releases" **Advanced** 1.5 h
- Monitoring models and AI apps in production **Advanced** 75 min
- Speed, cost and scaling **Advanced** 75 min

9

### Capstones

30–50 h

Churn prediction, demand forecasting and image inspection.

- Capstone 1: Customer churn prediction service 20–30 h
- Capstone 2: Store demand forecasting 20–30 h
- Capstone 3: Visual defect inspection 25–40 h

## AI engineer view roadmap

Focus on building products with language models: RAG, agents, evaluation and security. Learn enough ML to reason about models and data.

1

### Big picture and setup

4–6 h

Lifecycle and tools for Python and .NET.

- What AI, machine learning, deep learning and generative AI are **Beginner** 60 min
- How an AI or ML project runs from idea to production **Beginner** 75 min
- Set up your laptop for AI and ML in Python and .NET **Beginner** 1.5 h

2

### Enough data and ML

8–12 h

Data, how models learn, classification and evaluation.

- Data basics: types, features, labels and splits **Beginner** 60 min
- How a model learns: loss, training and overfitting **Beginner** 75 min
- Classification: picking a category **Beginner** 75 min
- Measuring models honestly **Intermediate** 1.5 h

3

### How LLMs work

6–10 h

Transformers, tokens, sampling and limits.

- Transformers and attention **Intermediate** 1.5 h
- How large language models work **Intermediate** 75 min
- Tokens and sampling settings **Intermediate** 60 min

4

### Calling models and prompting

6–8 h

APIs, structured output, prompt patterns.

- Calling an LLM from your application **Beginner** 75 min
- Prompt engineering that works in production **Beginner** 60 min

5

## Embeddings and RAG

10–14 h

Search by meaning and answer from your documents.

- Embeddings and vector search **Intermediate** 75 min
- RAG: answering from your own documents **Intermediate** 2 h
- Advanced RAG: chunking, hybrid search, re-ranking and evaluation **Advanced** 1.5 h

6

## Agents and multimodal apps

10–14 h

Tools, agents, frameworks, images and speech.

- Tools and agents with human approval **Advanced** 2 h
- AI frameworks and MCP: what to use in Python and .NET **Intermediate** 60 min
- Multimodal AI apps: images, documents and voice **Intermediate** 60 min

7

## Quality, security and responsibility

8–12 h

Evaluation, attacks and defences, fairness and privacy.

- Evaluating AI applications **Intermediate** 75 min
- Security for AI applications **Advanced** 75 min
- Responsible AI: fairness, privacy, transparency and oversight **Intermediate** 60 min

8

## Own and run models

10–16 h

Fine-tune, run locally, evaluate; serve and scale.

- Fine-tuning: SFT, LoRA, QLoRA and preference tuning **Advanced** 1.5 h
- Running language models on your own machine **Intermediate** 75 min
- Evaluating language models **Advanced** 60 min
- Serving a model as an API **Intermediate** 1.5 h
- Speed, cost and scaling **Advanced** 75 min
- Production architecture for AI systems **Expert** 2 h

9

**Capstones** **30–50 h**

RAG assistant, approval-based agent and a local private assistant.

- Capstone 4: Company knowledge assistant (RAG) 20–35 h
- Capstone 5: Order-returns agent with human approval 25–40 h
- Capstone 7: Private assistant running fully on a laptop 20–30 h
- Capstone 8: Fine-tune a small model with LoRA 20–30 h